

Final Report

Autonomous Spine-Leaf Data Center Architectures: A Hybrid Q-Learning
and BDI Approach for High-Performance Networking

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1. Introduction

Modern data centers are increasingly dominated by east-west traffic, driven by microservices, virtualization, and AI training clusters. This is addressed via non-blocking, low-latency, horizontally scalable solutions found in spine-leaf (Clos) topologies. However, under dynamic, bursty packet flows that generate hotspots, raise tail latency, and raise the possibility of Service-Level Agreement (SLA) violations, traditional protocols (OSPF, BGP/ECMP) falter (Molenaar, 2023).

This paper assesses a hybrid intelligent agent solution that combines a Belief-Desire-Intention (BDI) agent for deliberative governance and security with a Q-Learning reinforcement learning agent for stochastic traffic engineering. These agents communicate with real-time telemetry through a Model Context Protocol (MCP) server (Prometheus, Cisco DNA Center). The goal is to show how this hybrid design outperforms static routing in autonomous, adaptive orchestration (Andersen, 2025).

2. Aim and Objectives

The main goal is to assess how a hybrid system combining Q-Learning and BDI agents can manage spine-leaf networks more effectively. This system aims to improve data speed, reduce traffic jams, and follow rules better than traditional methods (Liu et al., 2025).

Objectives include:

- Define the PEAS (Performance measure, Environment, Actuators, Sensors) descriptors and compare the properties of reactive agents (e.g., Q-Learning) versus deliberative agents (e.g., BDI) in the context of spine-leaf architectures.
- Synthesize the existing literature on reinforcement learning-based traffic engineering in Software Defined Networking (SDN) and on BDI/cognitive agents in network management.
- Propose and evaluate a hybrid architecture mediated by Multiple Control Protocols (MCP) against traditional methods, highlighting any performance and security improvements.

- Quantitatively outline the expected outcomes, such as achieving more than 15% lower tail latency, increased link utilization, and enhanced auditable policy enforcement, based on conceptual insights and literature review.

These goals target measurable improvements in latency, throughput, and policy compliance (Liu et al., 2025).

3. Literature Review

3.1 Q-Learning for Dynamic Traffic Engineering

A temporal-difference reinforcement learning technique known as Q-Learning effectively addresses Markov Decision Processes without requiring explicit transition models. In this approach, the agent selects actions (such as path or spine selection), observes states (including link utilization, queue depths, and flow statistics), and receives rewards based on throughput, latency, and fairness in software-defined networking (SDN) environments, particularly in spine-leaf network architectures (Zhang et al., 2020).

By selectively rerouting important packet flows while preserving Equal-Cost Multi-Path (ECMP) routing for smaller flows, studies such as CFR-RL demonstrate significant advantages. These approaches reduce maximum link utilization and flow completion times. Additionally, Deep Q-Network variants outperform static hashing in fat-tree and Clos network fabrics, achieving load balancing that is independent of topology (Tosounidis et al., 2020).

3.2 BDI Architecture for Governance and Security

BDI agents uphold their intentions (committed plans), desires (goals like SLA maintenance, zero-trust compliance), and beliefs (the current world model from telemetry). In domains where pure RL runs the risk of dangerous exploration, such as fault management, anomaly response, and high-level policy enforcement, this deliberative reasoning excels (Rao & Georgeff, 2025).

BDI has been used to enable explainable, goal-directed behaviour in cybersecurity and telecommunications network management. Symbolic reasoning and learnt optimisation are combined in hybrid BDI-RL systems to create agents that are reliable and adaptable (Andersen, 2025).

3.3 Hybrid Integration via MCP

The Model Context Protocol (MCP) provides a secure and standardized connection between AI agents and external tools. An MCP server exposes Prometheus metrics, switch APIs, and policy engines as structured context, which allows for real-time state updates for Q-Learning agents and updated beliefs for BDI agents (Ehtesham et al., 2025).

This layered approach is supported by literature on Software-Defined Networking (SDN) knowledge planes and hybrid agent architectures, with BDI operating at the orchestration layer and reinforcement learning (RL) functioning at the reactive control plane (Ehtesham et al., 2025).

3.4 PEAS Descriptors

- **Performance:** Enforce security/QoS policies, reduce packet loss and tail latency, and maximise goodput.
- **Environment:** Variable east-west flows and failures in a stochastic spine-leaf fabric.
- **Actuators:** Set up SDN flow rules, reroute routes, and implement quarantine and ACLs.
- **Sensors:** MCP-fed telemetry (bandwidth, flows, logs, security events).

BDI enforces deterministic constraints, while Q-Learning controls stochastic optimisation.

4. Analysis of Findings

Q-Learning finds policies that intelligently reroute packet traffic, achieving better load distribution and lower congestion than hash-based ECMP, which often collides with huge flows. This hybrid model works better than conventional OSPF/BGP/ECMP. Link utilization and flow completion time improvements of 10–25% in simulated Clos fabrics have been reported in the literature (Cheng et al., 2024).

BDI helps prevent RL from violating invariants during the learning process. It does this by comparing suggested actions to high-level goals, such as "avoid paths through vulnerable segments" or "maintain redundancy." The observe-act loop is completed by the Model Comparison Protocol (MCP), which ensures that both agents operate using new, detailed data (Rao & Georgeff, 2025).

Strengths: Effective and understandable autonomy emerges from thoughtful oversight and adaptable responsiveness. The expansion of the horizontal framework enhances scalability. By implementing clear intentions, the security posture is strengthened.

Weaknesses and Trade-offs: In very large state spaces, Q-Learning may converge slowly and requires careful reward design, which can be improved using DQN and function approximation. Engineering efforts are necessary for managing BDI plan libraries. Despite the coordination overhead, it can be effectively controlled in a hierarchical manner (Tosounidis et al., 2020).

Faster convergence on smaller instances would be validated and exploration-exploitation issues highlighted by integration with module practical work (e.g., implementing tabular Q-Learning in a Mininet spine-leaf topology) (Tosounidis et al., 2020).

Overall, the hybrid architecture goes beyond single-paradigm solutions and directly supports the report's objectives by producing quantifiable improvements in governance and performance.

5. Conclusion

This study demonstrates that a robust system for autonomous management in spine-leaf data centers is produced by combining Q-Learning for path optimisation with BDI for policy management. This approach enhances security, adaptability, clarity, and performance while reducing congestion brought on by massive data volumes.

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